

Professional Summary

Artificial Intelligence and Data Science-focused Computer Science student skilled in **Machine Learning**, **Large Language Models (LLMs)**, **Retrieval-Augmented Generation (RAG)**, Natural Language Processing (NLP), Python, SQL, and Power BI. Experienced in developing **intelligent systems**, **recommendation engines**, and **predictive analytics solutions** while leveraging machine learning, semantic search, and data science to deliver scalable solutions, actionable business insights, and solve real-world problems.

Technical Skills

AI, ML & Deep Learning: Machine Learning, Neural Networks, NLP, LLMs, RAG, MLOps, TF-IDF, Scikit-learn, NumPy, Pandas

Data Analytics & BI: SQL, Power BI, Tableau, Excel, KPI Reporting, Dashboard Automation, Data Visualization

Programming & Analytics: Python, Statistical Analysis, EDA, Predictive Modeling

Data Engineering: ETL Pipelines, Data Validation, Data Warehousing, Data Modeling, Data Governance, MySQL, PostgreSQL

Advanced SQL: Joins, CTEs, Window Functions, Stored Procedures, Query Optimization

Professional Experience

Research Intern

May 2026 – Present

Deep Learning & Computer Vision Research

- Achieved **83% classification accuracy** and **79% macro F1-score** by designing and implementing a novel **multi-branch deep learning architecture** integrating spatial, contextual, interaction-aware, and wavelet-based frequency-domain feature learning, outperforming multiple benchmark PlantDoc research models.
- Achieved up to **99% accuracy** and **98% F1-score** on specific disease categories by developing custom **attention-based feature extraction**, cross-attention fusion, and spectral encoding modules for highly reliable recognition of fine-grained plant disease patterns.
- Achieved improved disease classification robustness by implementing **species-aware hierarchical classification**, **multi-scale feature representation learning**, and advanced model optimization techniques, resulting in reduced inter-species disease confusion.
- Conducted comprehensive model validation using **cross-validation**, ablation studies, and **Explainable AI** (Grad-CAM, LIME, SHAP) with **PyTorch**, delivering interpretable and research-grade computer vision solutions.

Microsoft Elevate & AICTE

Feb 2026 – Mar 2026

Power BI Data Analytics Intern

- Achieved **28% faster reporting** by developing **Power BI dashboards** and **automated KPI workflows** for SLA monitoring and operational analytics, resulting in improved decision-making efficiency.
- Achieved **99% reporting accuracy** by implementing **star-schema data models** and **SQL validation workflows** across **15,000+ records**, resulting in enhanced data quality and governance.
- Achieved improved business visibility by **optimizing SQL queries**, automating reporting processes, and delivering **actionable operational insights** through interactive analytics dashboards.

Technical Projects

Corrective RAG (CRAG) System

- Achieved **35% higher response relevance** by implementing a **Corrective RAG (CRAG)** pipeline using **LLMs**, **LangGraph**, and **vector databases**, improving retrieval accuracy for knowledge-intensive queries.
- Achieved **30% reduction in hallucinated responses** by developing a **retrieval evaluation framework** with document grading, query rewriting, and context validation, resulting in more reliable AI-generated outputs.
- Achieved **25% improvement in answer quality** by integrating query rewriting, knowledge refinement, and **semantic search**, enhancing context retrieval and user satisfaction.
- Achieved **40% better coverage** of out-of-domain queries by integrating **web search augmentation** and semantic retrieval workflows, resulting in enhanced answer completeness.

Recommendation System

- Achieved **24% higher recommendation precision** by designing a **hybrid model** integrating **collaborative filtering**, **content-based filtering**, and **matrix factorization**, improving personalization accuracy.
- Achieved **18% lower prediction error** by implementing cross-validation, **hyperparameter tuning**, and evaluation using **RMSE** and **Precision@K**, resulting in more reliable recommendations.
- Achieved **15% improvement** in early-stage recommendation relevance by mitigating the **cold-start problem** using **similarity-based initialization** and clustering techniques.
- Achieved **22% improvement in recommendation diversity** by implementing **similarity-based ranking** and **feature engineering** techniques, enhancing user engagement and content discovery.

Virtual Experience Programs

BCG – Data Science Advanced Simulation (Forage)

Jan 2026

- Performed **exploratory data analysis (EDA)** and **predictive churn modeling** using **Python** and **SQL-driven analytical workflows**.

Deloitte Australia – Data Analytics Job Simulation (Forage)

Oct 2025

- Executed data cleaning, **data visualization**, and business reporting on complex datasets for **strategic decision support**.

Education

VIT University

2023 – 2027

Bachelor of Technology in Computer Science & Engineering

CGPA: 8.4

Certifications

AWS Certified Machine Learning Engineer – Associate | AWS Certified Data Engineer – Associate | Oracle Cloud Infrastructure 2025

Certified Data Science Professional | AI & Data Science Professional – IIT Roorkee (iHUB)

DevOps and Cloud Foundations – IBM | Introduction to Data Science – CISOC